

Revised Generation Interconnection System Impact Study Report

For Queue Project Q17-0803

Fenton 100 kV
September 03, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0803. The Interconnection Customer proposes a 31.6 MW solar photovoltaic facility interconnecting at the Fenton 100 kV substation in the Michigan area. The request was received on May 27, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 336 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 13 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,270,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: CALDWELL – PINEY CREEK PROJECT 345kV
Contingency: HILLTOP – PINEY CREEK PROJECT 500kV
Loading: 103.6% (Rating: 2,276 MVA)
DFAX: 8.7%
Pre-project loading: 103.5%
2. Facility: MONTOUR STATION – TALENENERGY MARTINS CREEK LLC WILLIAMSPORT SUBSTATION 345kV
Contingency: ORCHARD RD – FT HUMBUG 500kV
Loading: 102.6% (Rating: 1,762 MVA)
DFAX: 5.1%
Pre-project loading: 102.5%
3. Facility: FISHKILL PLAINS – SIMPSON STUART TAP 345kV
Contingency: HAMPTON RD – MEDORA 345kV
Loading: 102.6% (Rating: 1,859 MVA)
DFAX: 6.0%
Pre-project loading: 102.5%
4. Facility: SHAWVILLE STA – BLACKROCK 345kV
Contingency: CARBON CTR 500-345 kV Transformer Bank
Loading: 102.5% (Rating: 2,111 MVA)
DFAX: 7.3%
Pre-project loading: 102.4%
5. Facility: CALDWELL – CARBON CTR 345kV
Contingency: BLACKROCK – SHAWVILLE STA 345kV
Loading: 102.3% (Rating: 1,506 MVA)
DFAX: 5.2%
Pre-project loading: 102.2%

6. Facility: GALION GENERATING STATION – MYRTLE POINT 230kV
Contingency: TREGO – BEAVER 500kV
Loading: 101.8% (Rating: 1,011 MVA)
DFAX: 7.4%
Pre-project loading: 101.5%
7. Facility: TALENENERGY MARTINS CRK LLC WILLIAMSPORT – HILLTOP 500kV line
Contingency: HILLTOP 500/345 kV TX
Loading: 101.7% (Rating: 2,941 MVA)
DFAX: 9.5%
Pre-project loading: 101.6%
8. Facility: NEW DEPARTURE – ADAMS 230kV Line
Contingency: TREGO – BEAVER 500kV Line
Loading: 101.4% (Rating: 864 MVA)
DFAX: 7.8%
Pre-project loading: 101.1%
9. Facility: MASON ST 500 kV/230 kV Transformer Bank
Contingency: TREGO – BEAVER 500kV Line
Loading: 101.2% (Rating: 1,308 MVA)
DFAX: 11.8%
Pre-project loading: 100.9%
10. Facility: EUFAULA – BEAVER 500kV
Contingency: POST ROCK – NEWPORT TIE 230kV
Loading: 101.2% (Rating: 3,064 MVA)
DFAX: 28.4%
Pre-project loading: 100.9%
11. Facility: HILLTOP – PINEY CREEK PROJECT 500kV
Contingency: PINEY CREEK PROJECT – CALDWELL 345kV
Loading: 100.6% (Rating: 2,934 MVA)
DFAX: 11.1%
Pre-project loading: 100.4%
12. Facility: HIGHGATE – HERALD-TWIN CITIES 230kV
Contingency: MASON ST SUBSTATION 500/230kV Transformer
Loading: 100.5% (Rating: 857 MVA)
DFAX: 6.9%
Pre-project loading: 100.2%
13. Facility: TALENENERGY MARTINS CRK LLC WILLIAMSPORT 500/345 kV Transformer
Contingency: HILLTOP 500-345 kV TX
Loading: 100.1% (Rating: 2,280 MVA)
DFAX: 6.1%
Pre-project loading: 100.0%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
PNEY CRK PROJECT – CALDWELL 345kV Line	Reconductor line	\$27,615,000	0% (\$127,000)
MONTOUR STATION – TALENENERGY MARTINS CREEK LLC WILLIAMSPORT SUBSTATION 345kV	Reconductor line	\$21,987,000	1% (\$185,000)
SIMPSON STUART TAP – FISHKILL PLAINS 345kV Line	Reconductor line	\$13,018,000	0% (\$60,000)
SHAWVILLE STA – BLACKROCK 345kV	Reconductor line	\$18,073,000	0% (\$83,000)
CALDWELL – CARBON CENTER 345kV	Reconductor line	\$19,637,000	0% (\$90,000)
GALION GENERATING STATION – MYRTLE POINT 230kV	Reconductor line	\$12,036,000	0% (\$56,000)
TALENENERGY MARTINS CRK LLC WILLIAMSPORT – HILLTOP 500kV line	Reconductor line	\$28,792,000	0% (\$132,000)
NEW DEPARTURE – ADAMS 230kV Line	Reconductor line	\$10,838,000	0% (\$51,000)
MASON ST 500 kV/230 kV Transformer Bank	Replace transformer with higher-rated unit	\$43,383,000	0% (\$205,000)
BEAVER – EUFAULA 500kV L	Reconductor line	\$18,360,000	0% (\$84,000)
HILLTOP – PINEY CRK PROJECT SUBSTATION 500kV	Reconductor line	\$224,763,000	0% (\$1,031,000)
HIGHGATE – HERALD-TWIN CITIES 230kV line	Reconductor line	\$9,406,000	0% (\$44,000)
TALENENERGY MARTINS CRK LLC WILLIAMSPORT 500/345kV Transformer	Replace transformer with higher-rated unit	\$26,516,000	0% (\$122,000)
	Total allocated to project		\$2,270,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.